

PROJECT PROPOSAL

Interpretable Sentiment Analysis of Customer Reviews Using Deep Learning

IE 7615 Deep Learning for AI | Wiktor Lasek

Problem Definition

Millions of customer reviews are posted online every day across platforms such as Amazon, Yelp, and Google. For businesses, these reviews represent an invaluable source of feedback about their products and services. However, manually reading and analysing thousands of reviews is impossible at scale. A small business owner with 500 reviews, a product manager tracking a new launch, or a customer service team monitoring complaints – all face the same challenge – understanding what customers truly think, quickly and accurately.

Traditional approaches such as rule-based classifiers and bag-of-words models can label reviews as positive or negative, but they will fail to capture context, sarcasm, and nuanced language [3]. More critically, they cannot explain why a review was classified a certain way. A business needs to know whether the review is negative or positive and whether customers are complaining about delivery, product quality, or customer service.

This project addresses both challenges by building a progressive deep learning pipeline that classifies customer review sentiment and visually explains which words and phrases drove each prediction – using attention visualization [4]. This interpretability makes the system genuinely actionable for real business decision-making.

Data Source

This project uses two datasets:

Amazon Customer Review (train.ft.csv) [1]	Kaggle	~1,000,000 reviews	Training all models.
Amazon Customer Reviews (test.ft.csv) [1]	Kaggle	~400,000 reviews	Standard model evaluation.
Hospital Patient Reviews [2]	Kaggle	~1,000 reviews	Cross-domain generalization test, evaluating whether the best model performs on an entirely different domain.

The Amazon dataset is pre-split into training and test sets, making standard evaluation straightforward. Both datasets are publicly available and raise no ethical concerns. Text preprocessing includes tokenization, stop-word removal, lowercasing and sequence padding.

Proposed Approach

The goal is to build a system that reads customers reviews and automatically detects whether the experience was positive or negative. To do this, four models will be built and compared, starting simple and getting more advanced:

1. **Logistic Regression (Baseline)** The simplest model – it counts word frequencies using TF-IDF and makes a prediction based on that. This gives us starting score to beat.
2. **Feedforward Neural Network** A basic neural network that learns patterns beyond simple word counts. Slightly smarter than the baseline.
3. **LSTM (Long Short-Term Memory)** [3] A more advanced model that reads the review word by word and remembers the context – understanding that “not good” is negative, not positive.
4. **DistilBERT (Transformer)** [5] A pre-trained transformer model that deeply understands English text. It will be fine-tuned on the Amazon reviews dataset for sentiment classification and will generate attention heatmaps showing which words most influenced each prediction [6].

All models will be evaluated on the Amazon test set (400000 reviews). The best performing model will be tested on the hospital reviews dataset to verify cross-domain

generalization using Accuracy, F1-Score and ROC-AUC. All experiments will be run in Python.

Timeline

1	May 11-17	Define the problem	Literature review, download and explore datasets, define labelling strategy and success metrics
2	May 18-24	Specify requirements and brainstorm	Data preprocessing, TF-IDF feature extraction, compare model architectures, set up Colab environments
3	May 25-31	Choose best solution and develop	Implement Logistic Regression and Feedforward NN baselines
4	Jun 1-7	Build prototype	Implement and train LSTM model with GloVe embeddings
5	Jun 8-14	Test and redesign	Fine-tune DistilBERT, generate attention heatmaps, cross-dataset testing on hospital reviews, compare all models
6	Jun 15-21	Communicate results	Finalize analysis, write final report

References

- [1] Amazon Customer Reviews Dataset. (2023). Kaggle. <https://www.kaggle.com/datasets/bittlingmayer/amazonreviews>
- [2] Hospital Reviews Dataset. (2023). Kaggle. <https://www.kaggle.com/datasets/junaid6731/hospital-reviews-dataset>
- [3] Bishop, C. M. (n.d.). *Pattern Recognition and Machine Learning*. Retrieved May 2025, from <https://www.bishopbook.com>
- [4] Devlin, J., Chang, M., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding. *Proceedings of NAACL-HLT 2019*. <https://arxiv.org/abs/1810.04805>
- [5] Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. <https://arxiv.org/abs/1910.01108>
- [6] Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30. <https://arxiv.org/abs/1706.03762>